SAP Business Data Cloud Analytics

Working Capital and Company Code Clustering Dashboard

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Program

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Introduction

This capstone project is based on SAP Business Data Cloud and focuses on analyzing working capital and company code performance using data-driven techniques. The project utilizes SAP Analytics Cloud, SAP Datasphere, and SAP Databricks to integrate, model, and visualize enterprise financial data.

The main objective of this project is to develop a clustering dashboard that helps in identifying patterns in working capital and cash flow across different company codes. By applying data modeling and visualization techniques, the project enables better financial analysis, improved decision-making, and efficient monitoring of business performance.

This work aligns with the Record-to-Report (R2R) domain, as it supports financial reporting, working capital analysis, and data-driven insights for business users.

Problem Statement

In large organizations, financial data is often distributed across multiple systems and company codes, making it difficult for finance teams to obtain a unified view of liquidity, cash flow, receivables, payables, and overall working capital performance. This lack of integration leads to challenges in monitoring financial health and making timely decisions.

Additionally, business users require insights into how different company codes behave in relation to each other. However, traditional reporting methods primarily present numerical data and trends, which do not effectively highlight patterns or groupings based on financial behavior. This limits the ability to perform benchmarking, strategic comparison, and efficient financial planning.

Therefore, there is a need for a solution that not only consolidates financial data but also enables intuitive visualization and segmentation of company codes to improve analysis and decision-making.

Objectives

The main objective of this project is to develop an SAP Business Data Cloud–based analytics solution that enhances working capital analysis through clustering-based visualization of company codes.

The specific objectives are:

- To understand the SAP Business Data Cloud architecture and its integrated components.
- To work with intelligent finance application content in SAP Analytics Cloud.
- To integrate clustering outputs (cluster labels and TSNE coordinates) into the analytic model.
- To design a scatterplot-based dashboard for company code segmentation and analysis.

Tech Stack

The project utilizes the SAP Business Data Cloud ecosystem along with associated tools and technologies. SAP Analytics Cloud is used for dashboard creation and data visualization. SAP Datasphere is used for data integration and modeling, while SAP Databricks is used for performing clustering and advanced data processing. SAP S/4HANA serves as the primary source of financial data.

Additionally, clustering techniques such as TSNE are used for dimensionality reduction and visualization of company code patterns.

System Architecture

The SAP Business Data Cloud architecture integrates multiple components such as SAP S/4HANA, SAP Datasphere, SAP Databricks, and SAP Analytics Cloud. SAP S/4HANA serves as the primary source of financial data, while SAP Datasphere is used for data integration and modeling. SAP Databricks enables advanced data processing and clustering, and SAP Analytics Cloud is used for visualization and dashboard creation.

This integrated architecture enables seamless data flow and supports advanced analytics for business decision-making.

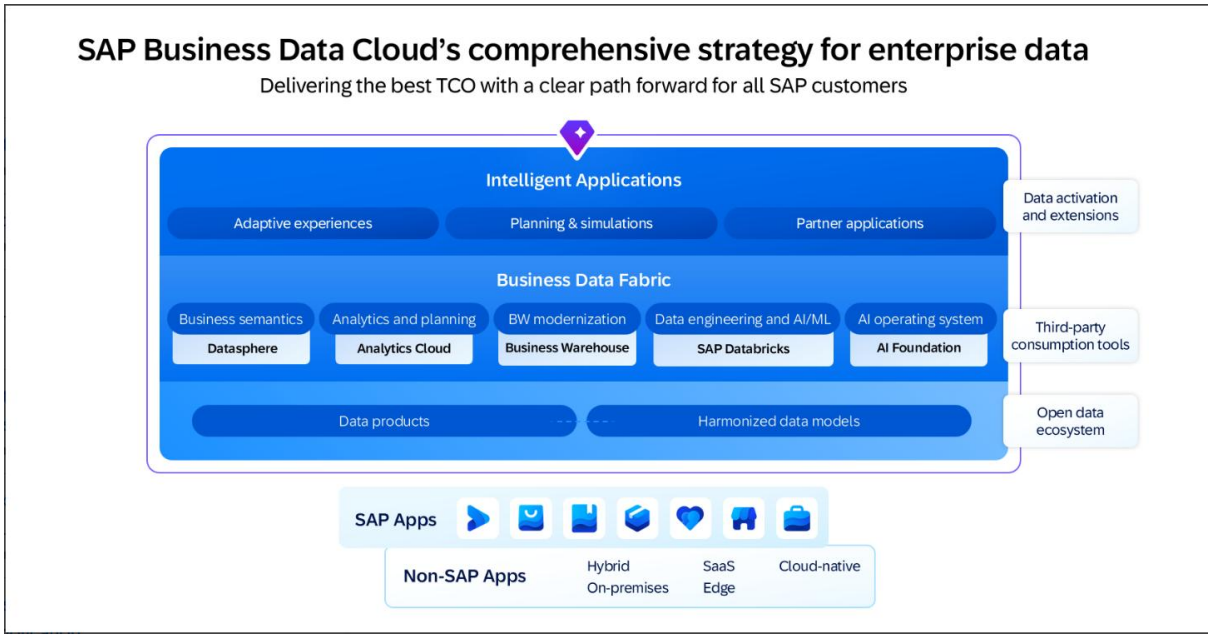

Figure 1: SAP Business Data Cloud Architecture

Solution and Implementation

The implementation of this project involves the integration of multiple SAP Business Data Cloud components to enhance working capital analysis through clustering-based visualization. The process includes accessing SAP Analytics Cloud, working with intelligent application content, integrating data models from SAP Datasphere, and utilizing clustering outputs generated from SAP Databricks.

Step 1: Accessing SAP Analytics Cloud

The implementation begins by accessing SAP Analytics Cloud, which serves as the primary platform for data visualization and dashboard creation. The interface provides features such as Stories, Data Analyzer, Modeler, and Files, enabling users to manage and analyze enterprise data efficiently.

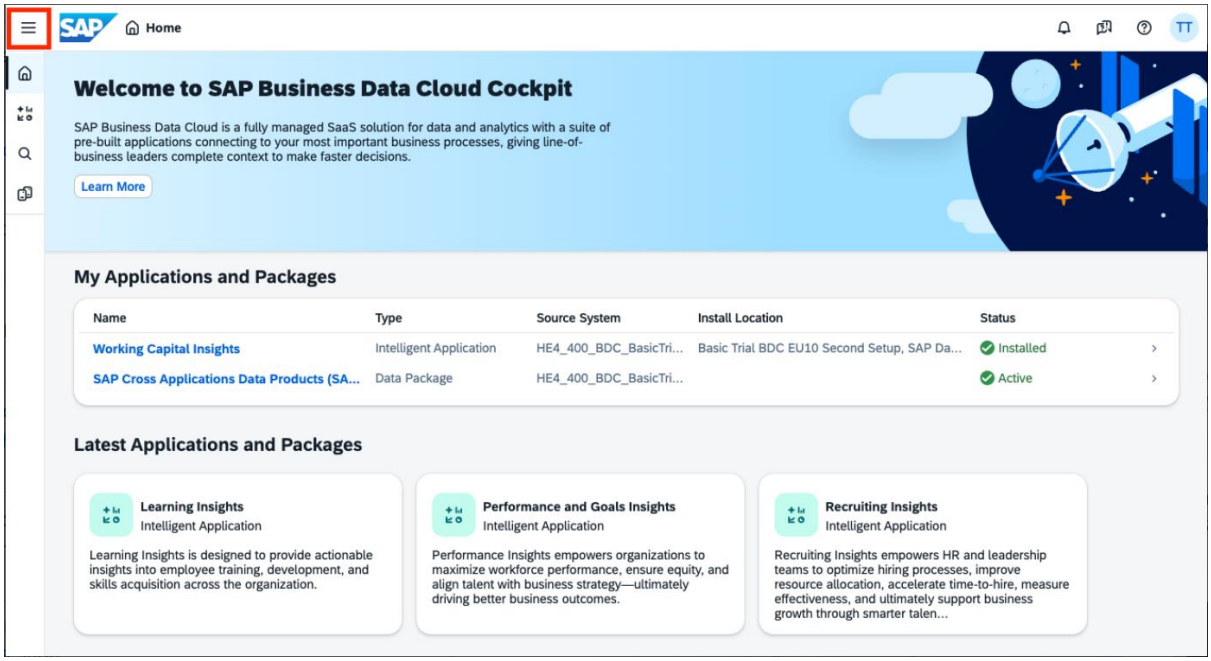

Figure 2: SAP Analytics Cloud Home Interface

Step 2: Accessing Intelligent Applications

After logging into SAP Analytics Cloud, the intelligent application content is accessed from the shared folders. The Working Capital Insights story is selected from the SAP S/4HANA finance content, which serves as the base for further customization.

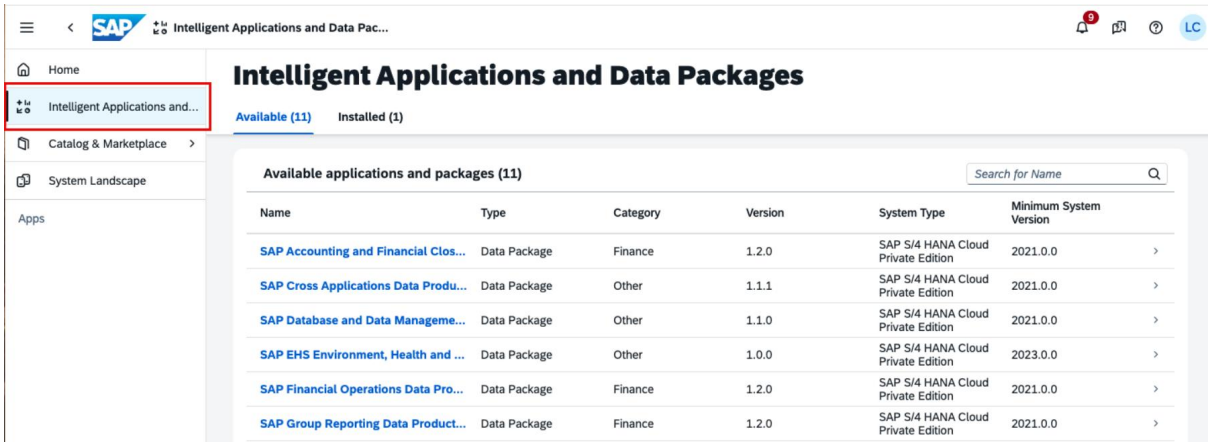

Figure 3: Intelligent Applications in SAP Analytics Cloud

Step 3: Copying and Editing the Story

The selected Working Capital story is copied into the personal workspace to allow customization without affecting the original content. The copied story is then opened in edit mode, where modifications are made to enhance the dashboard with additional analytical features.

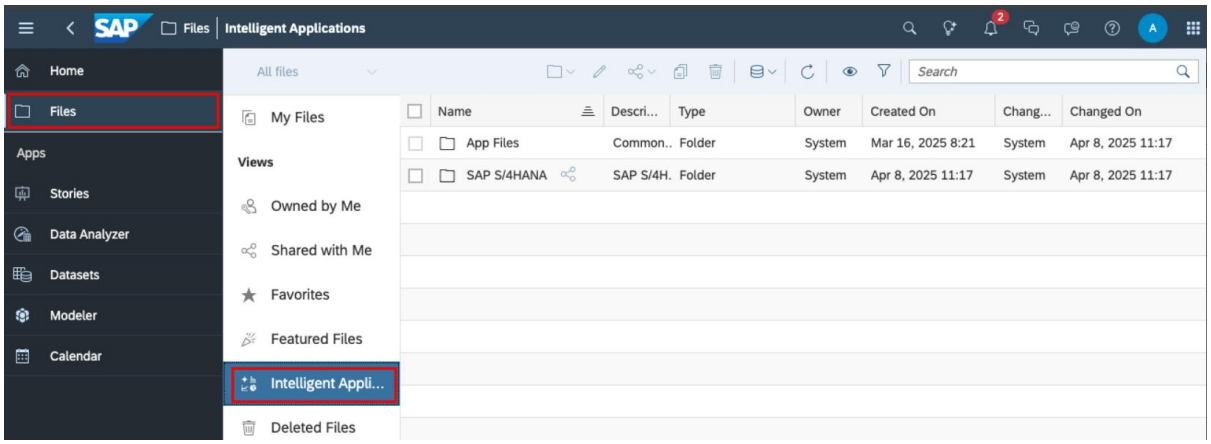

Figure 4: Accessing Intelligent Applications in SAP Analytics Cloud

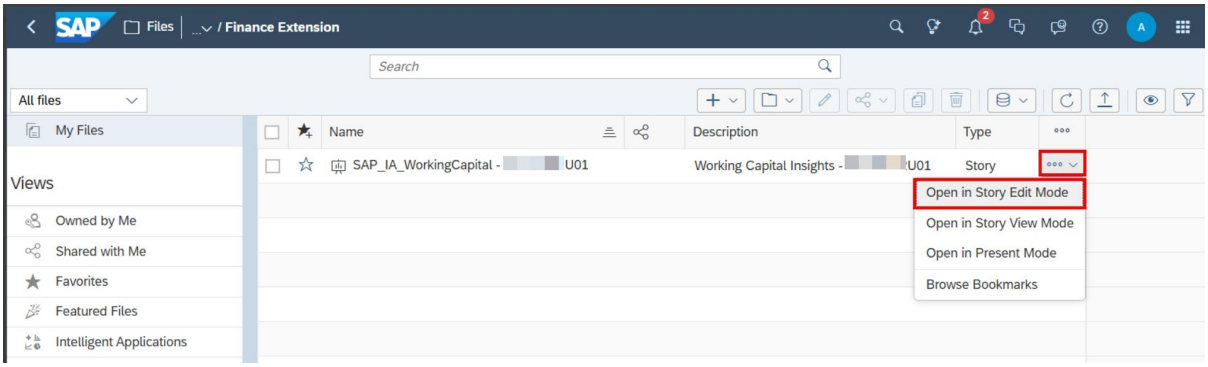

Figure 5: Opening Working Capital Story in Edit Mode

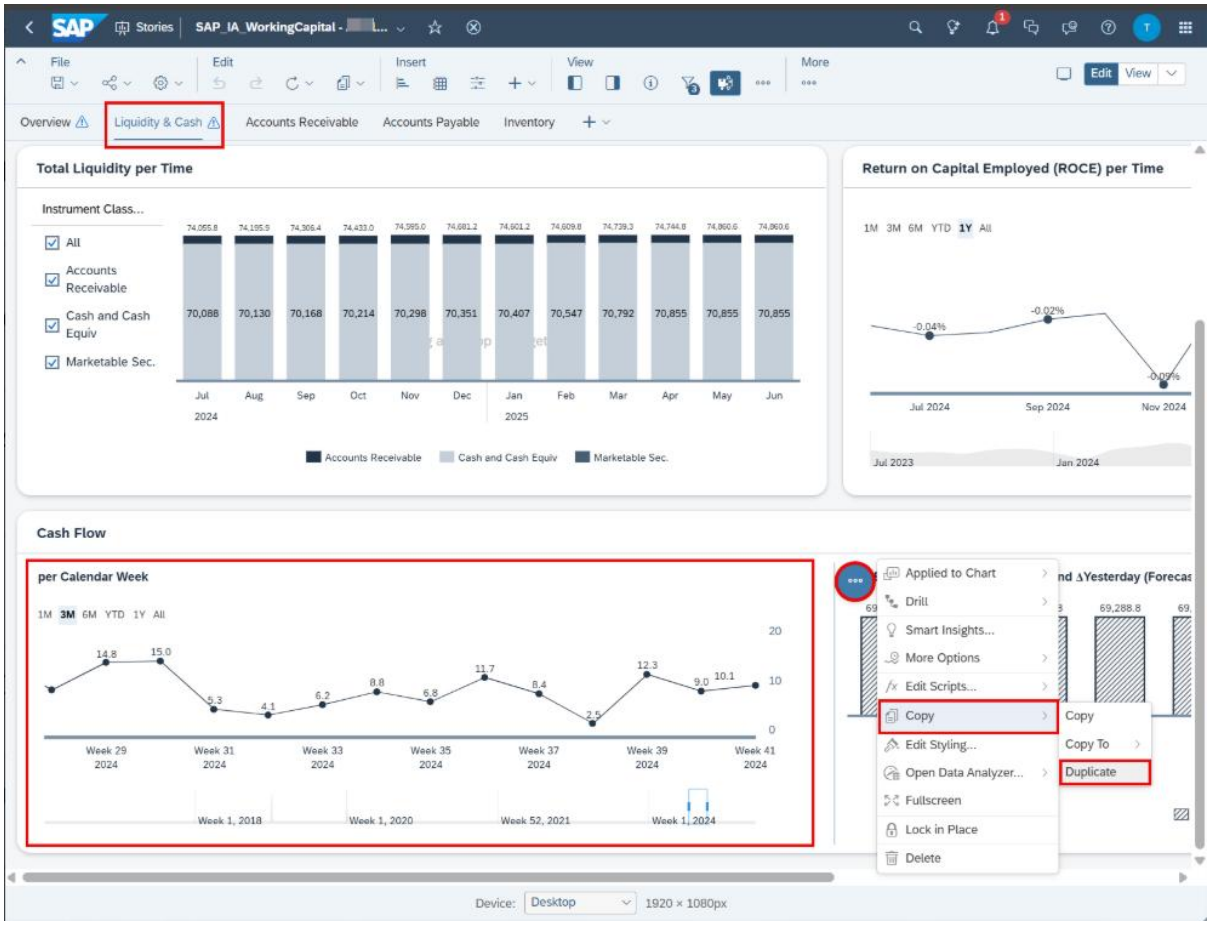

Figure 6: Editing and Customizing the Working Capital Dashboard

Step 4: Integration with SAP Datasphere

SAP Datasphere is used for data integration and modeling. The analytic model is extended by including additional attributes such as company cluster labels and

TSNE coordinate values. These attributes are essential for visualizing clustering results in the dashboard.

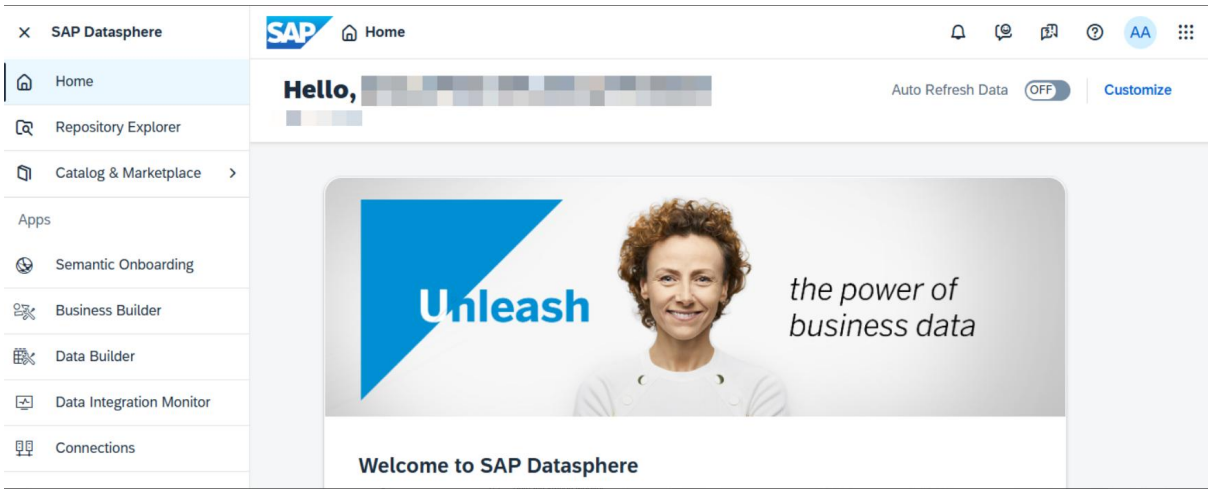

Figure 7: SAP Datasphere Home Screen

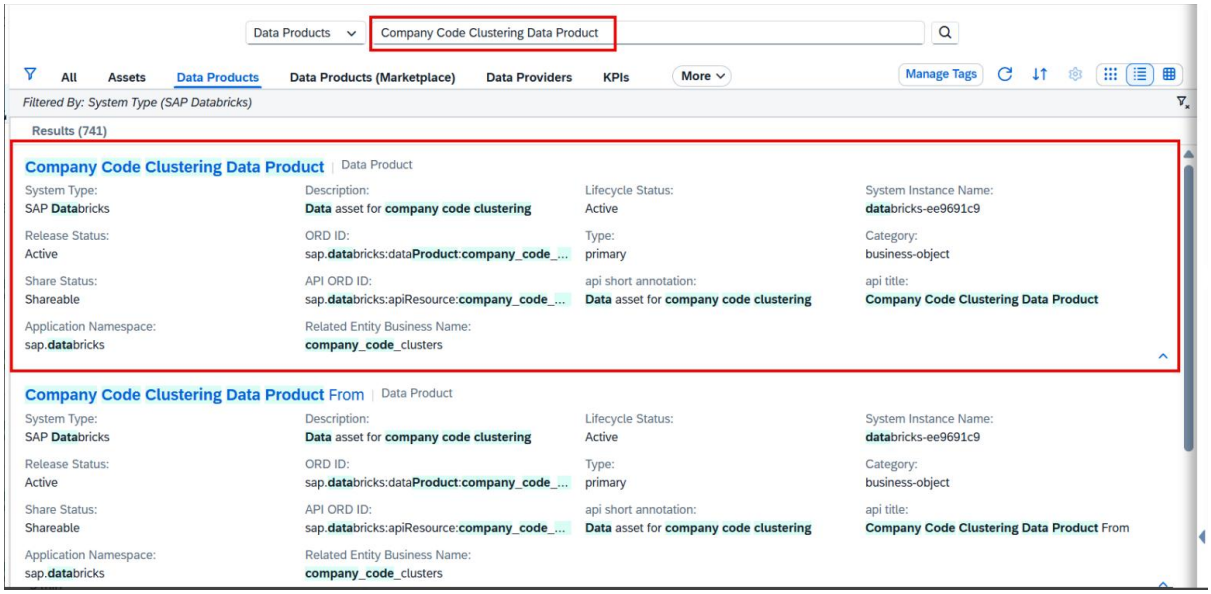

Figure 8: Company Code Clustering Data Product in SAP Datasphere

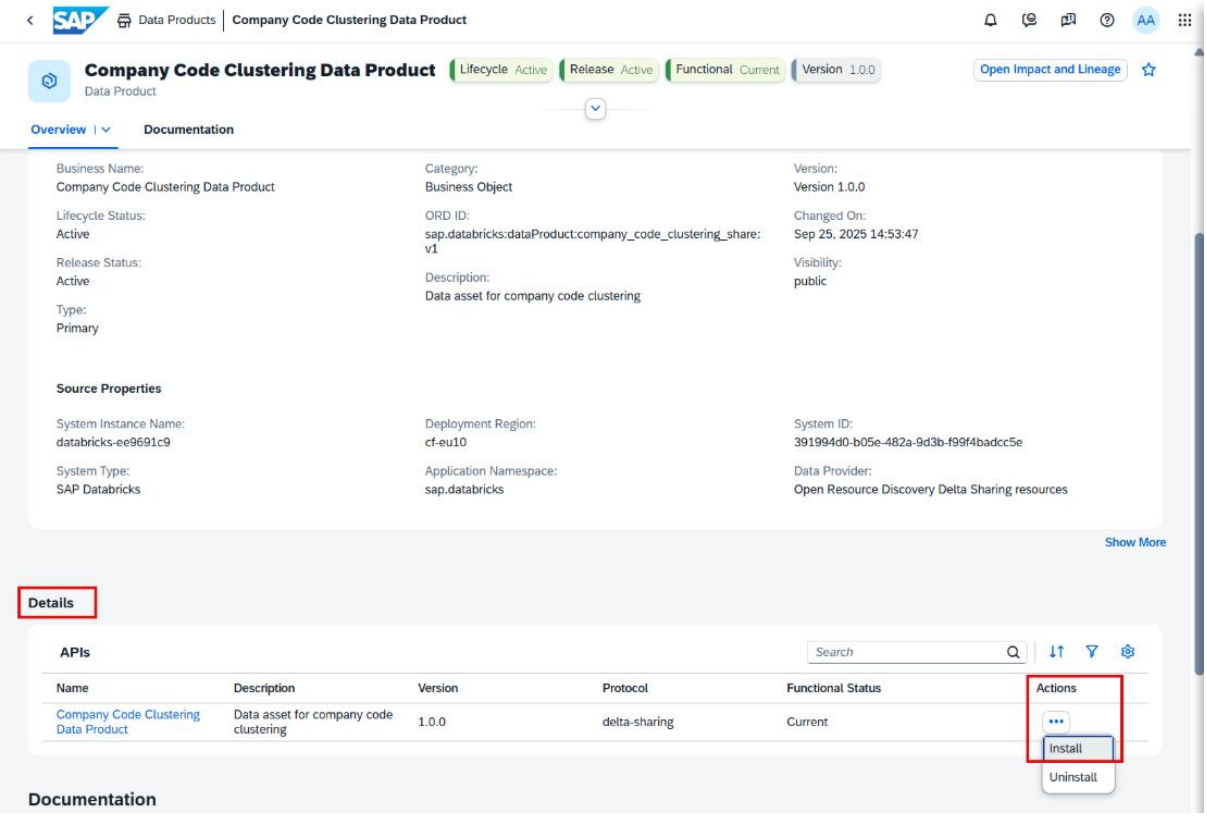

Figure 9: Import Entities Screen in SAP Datasphere

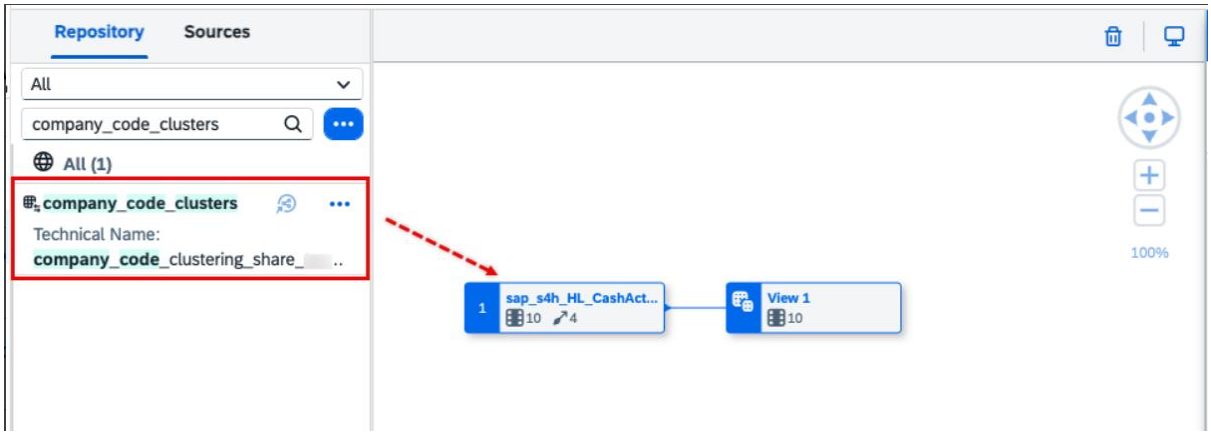

Figure 10: Data Builder Join Node in SAP Datasphere

Step 5: Clustering using SAP Databricks

SAP Databricks is utilized to perform clustering on company code data. The clustering process generates outputs such as cluster labels and coordinate values, which are then integrated into the SAP Datasphere model. This enables advanced data processing and segmentation of company codes based on financial behavior.

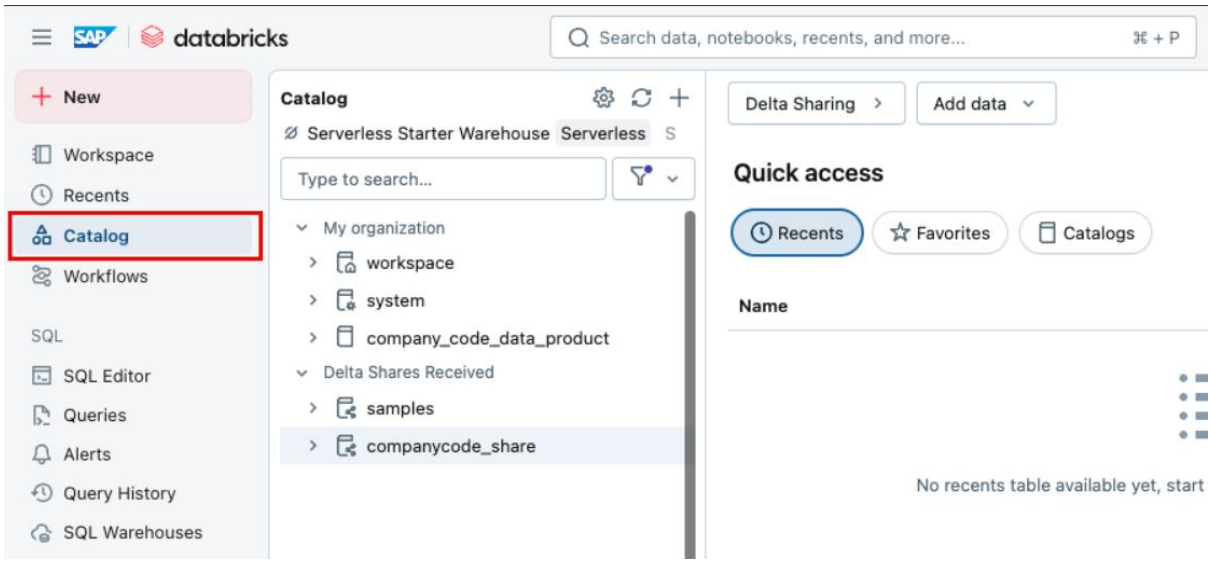

Figure 11: SAP Databricks Catalog

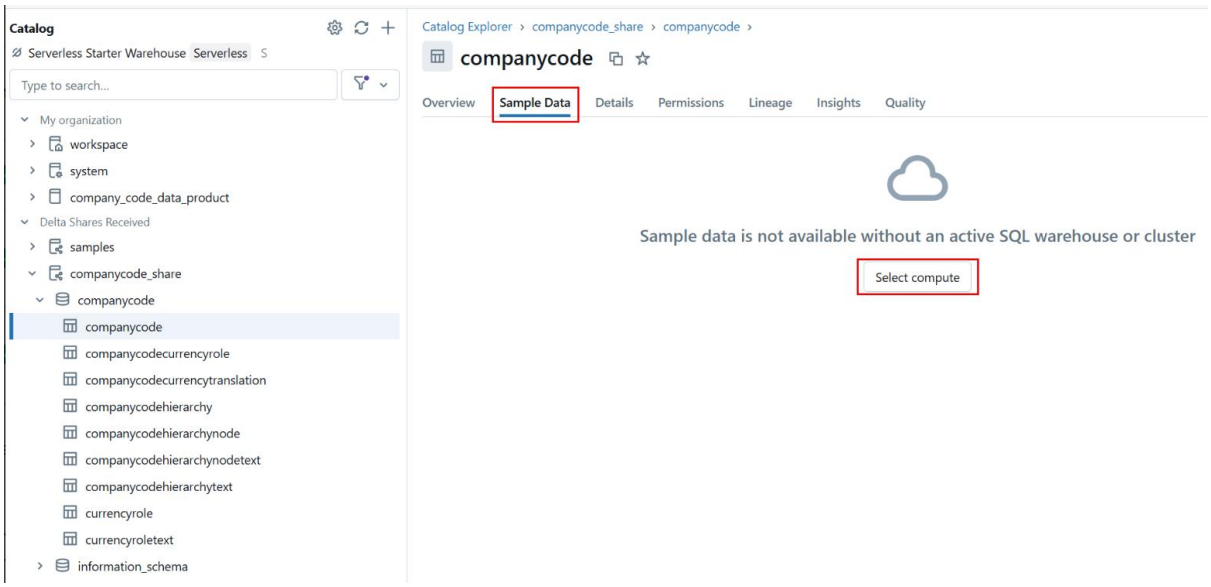

Figure 12: Company Code Tables in Databricks

Step 6: Scatterplot Configuration in SAP Analytics Cloud

The final step involves configuring a scatterplot in SAP Analytics Cloud using TSNE X and TSNE Y as axes, Company Code as the dimension, and Country/Region as the color attribute. This visualization enables clear identification of clusters and helps users analyze similarities and differences among company codes.

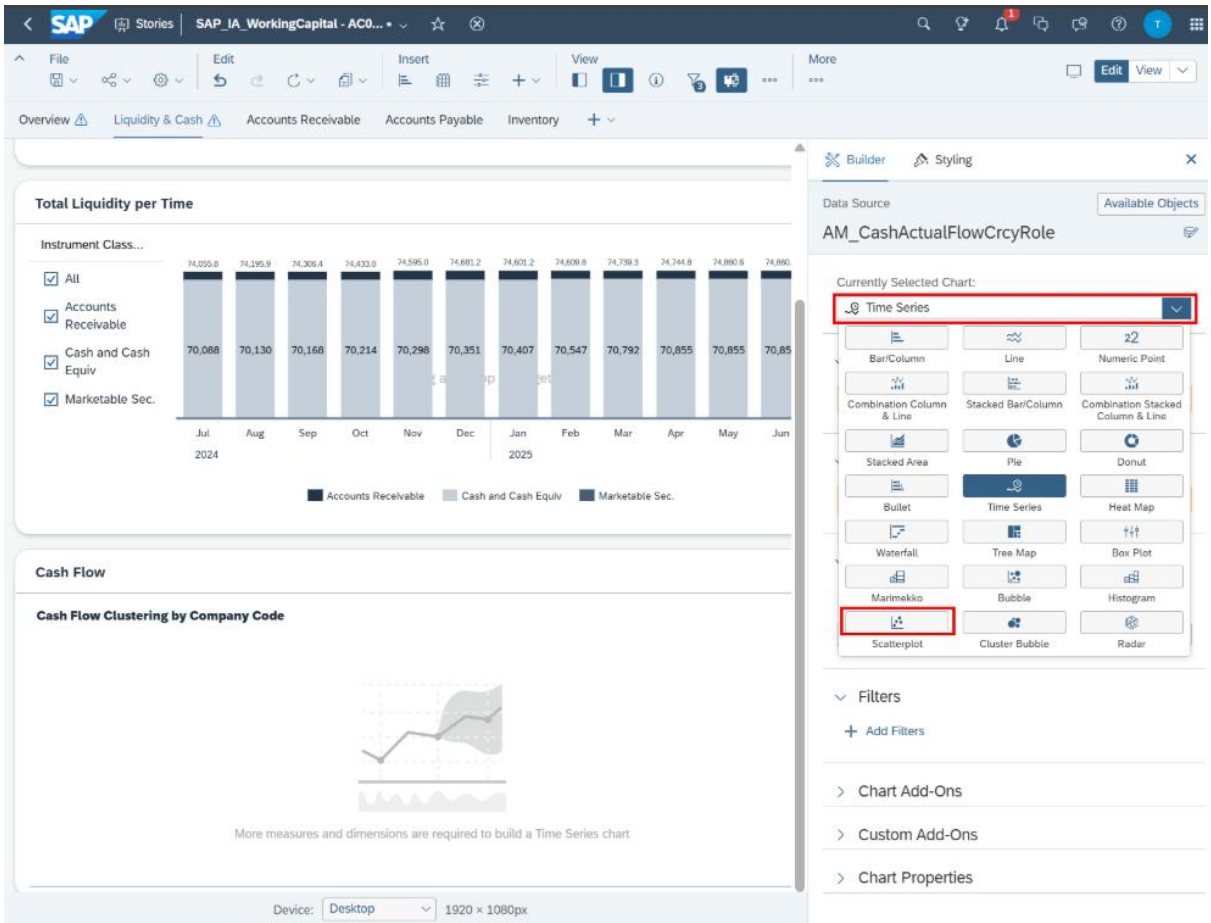

Figure 13: Selecting Scatterplot Visualization in SAP Analytics Cloud

This step involves selecting the scatterplot chart type in SAP Analytics Cloud for visualizing clustering results. The scatterplot enables representation of company codes as data points, which helps in identifying patterns and relationships in financial data.

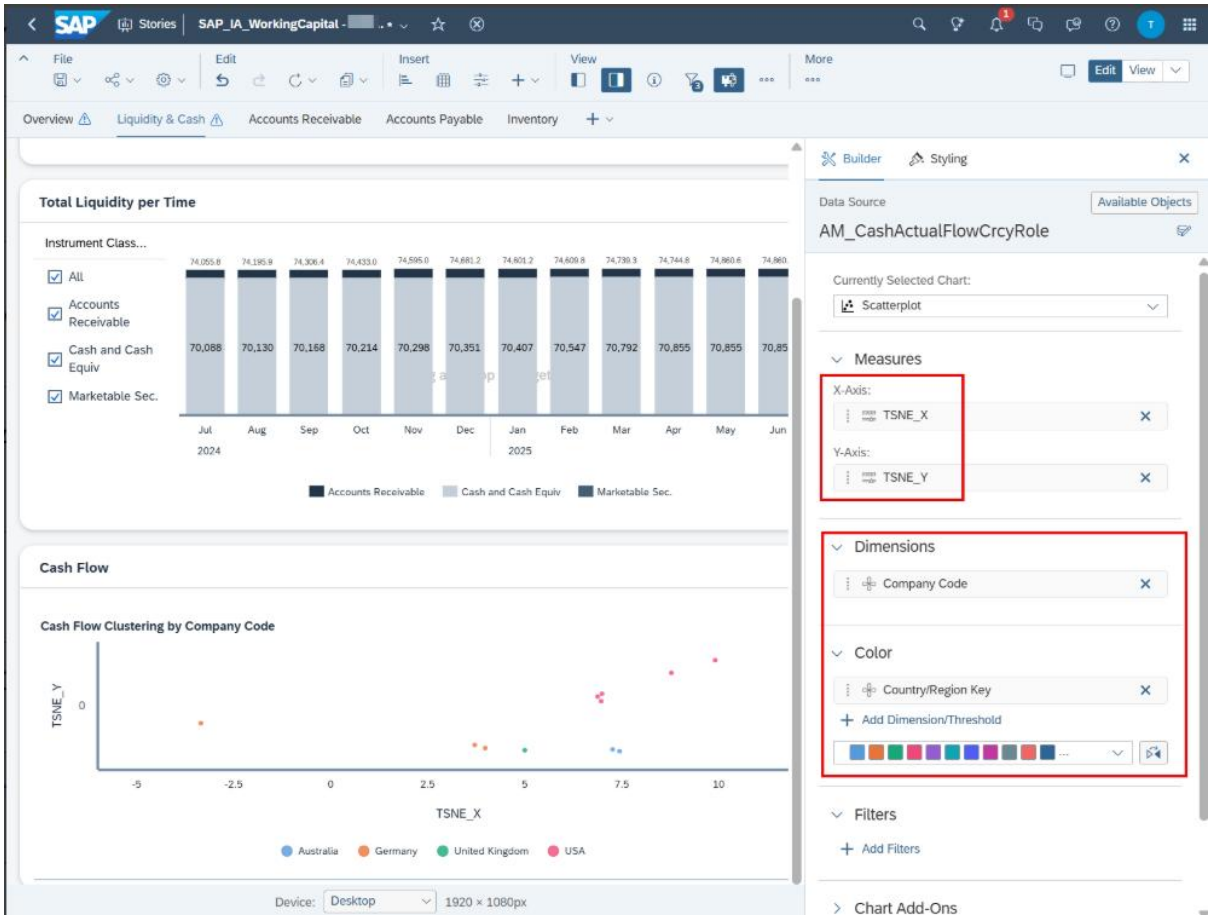

Figure 14: Configuring Scatterplot using TSNE X and TSNE Y

This step involves configuring the scatterplot by assigning TSNE X and TSNE Y as the horizontal and vertical axes respectively. The Company Code is used as the dimension, while Country/Region is applied as a color attribute to differentiate data points. This configuration enables clear visualization of clusters, helping users identify similarities and patterns in financial behavior across different company codes.

Unique Features

- Integration of multiple SAP components including SAP Analytics Cloud, SAP Datasphere, and SAP Databricks.
- Use of clustering techniques to enhance financial data analysis.
- Visualization of company code clusters using scatterplot for better interpretation.
- Interactive dashboard enabling comparative analysis across different company codes.
- Improved decision-making through data-driven insights and visual analytics.

Dashboard and Results

The final dashboard developed in SAP Analytics Cloud provides a comprehensive view of working capital metrics along with clustering-based visualization of company codes. It integrates financial data with advanced analytics to enable better understanding of business performance.

The scatterplot visualization plays a key role in this dashboard by representing company codes as data points based on TSNE X and TSNE Y values. Each point is color-coded based on country or region, allowing users to easily distinguish between different groups. This helps in identifying clusters of company codes with similar financial behavior.

The dashboard enables users to analyze working capital trends, compare performance across company codes, and detect patterns that are not easily visible in traditional reports. This improves decision-making by providing clear and interactive insights into financial data.

Overall, the solution enhances financial analysis by combining data modeling, clustering techniques, and visualization, making it easier for business users to interpret complex data and take informed decisions.

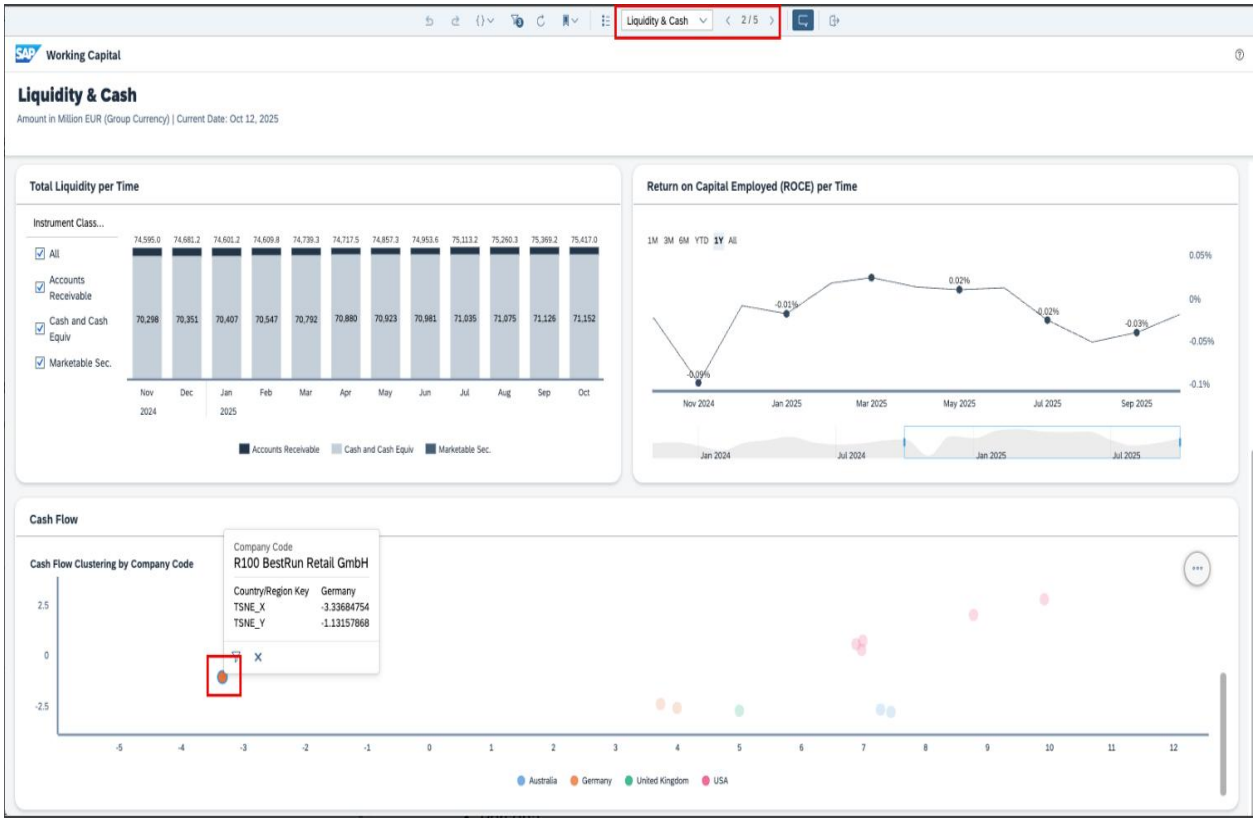

Figure 15: Final Dashboard Showing Working Capital Analysis and Company Code Clustering

This dashboard supports business users in performing benchmarking and strategic analysis across company codes. It also enables identification of outliers and optimization opportunities in working capital management.

Conclusion

This project demonstrates the effective use of SAP Business Data Cloud in enhancing financial analysis through data integration, clustering, and visualization techniques. By combining SAP Analytics Cloud, SAP Datasphere, and SAP Databricks, the solution provides a unified and interactive platform for analyzing working capital and company code performance.

The implementation of scatterplot-based clustering enables better identification of patterns and relationships among company codes, which are not easily visible in traditional reporting methods. This improves decision-making by providing clear and data-driven insights to business users.

Overall, the project highlights the importance of advanced analytics and visualization in modern enterprise systems and showcases how SAP Business Data Cloud can be leveraged to drive efficient and informed financial analysis.

Future Scope

The current project can be further enhanced by incorporating advanced analytical techniques such as predictive modeling and machine learning algorithms to forecast working capital trends and financial performance. Integration of real-time data processing can also improve the responsiveness of the dashboard, enabling more dynamic and up-to-date insights.

Additionally, the solution can be extended by including more financial KPIs and expanding the analysis to other business domains such as procurement, sales, and supply chain. Enhancing user interactivity through advanced filtering, drill-down features, and AI-driven insights can further improve decision-making capabilities.

References

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